



CALCULO INTEGRAL BMA02

semana 4

Método de integración:
De racionales en seno y coseno

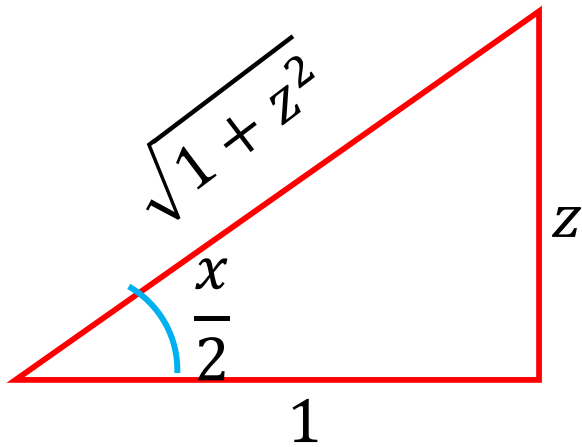
$$I = \int \frac{2+3\cos x}{1+\sin x - \cos x} dx$$





Método de Integración de racionales en seno y coseno

La sustitución recomendada es: $z = \tan \left[\frac{x}{2} \right]$



$$\operatorname{sen}\left(\frac{x}{2}\right) = \frac{z}{\sqrt{1+z^2}} \quad \cos\left(\frac{x}{2}\right) = \frac{1}{\sqrt{1+z^2}}$$

$$\operatorname{sen} x = 2 \operatorname{sen}\left(\frac{x}{2}\right) \cos\left(\frac{x}{2}\right) = \frac{2z}{1+z^2}$$

$$\cos x = \cos^2\left(\frac{x}{2}\right) - \operatorname{sen}^2\left(\frac{x}{2}\right) = \frac{1-z^2}{1+z^2}$$

$$dz = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) dx \quad dx = \frac{2dz}{1+z^2}$$



Método de Integración de racionales en seno y coseno

Ejemplo: Calcule $I = \int \frac{1 + \operatorname{sen} x + 2 \cos x}{1 - \cos x + \operatorname{sen} x} dx$

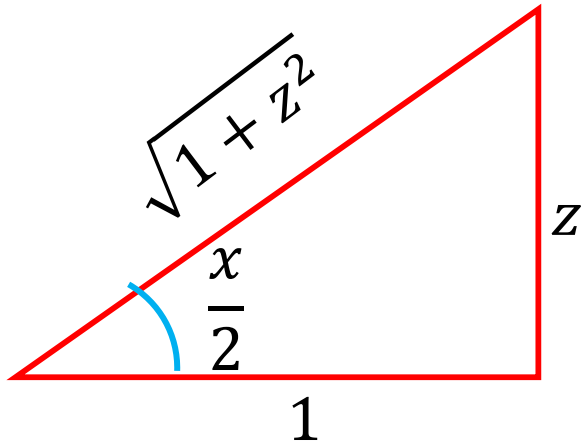
$$\operatorname{sen}\left(\frac{x}{2}\right) = \frac{z}{\sqrt{1+z^2}}$$

$$\cos x = \frac{1-z^2}{1+z^2}$$

$$\cos\left(\frac{x}{2}\right) = \frac{1}{\sqrt{1+z^2}}$$

$$dz = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) dx$$

$$I = \int \frac{1 + \frac{2z}{1+z^2} + 2 \frac{1-z^2}{1+z^2}}{1 - \frac{1-z^2}{1+z^2} + \frac{2z}{1+z^2}} dx$$



$$dx = \frac{2dz}{1+z^2}$$

$$I = \int \frac{1+z^2+2z+2-2z^2}{1+z^2-1+z^2+2z} dx$$

$$I = \int \frac{3+2z-z^2}{2z^2+2z} \frac{2dz}{1+z^2} = - \int \frac{z^2-2z-3}{z^2+z} \frac{dz}{1+z^2} = - \int \frac{z-3}{z(1+z^2)} dz$$

$$I = 3 \int \frac{dz}{z} - \frac{3}{2} \int \frac{2z}{(1+z^2)} dz + \int \frac{1}{(1+z^2)} dz$$

$$I = 3 \operatorname{Ln}\left(\operatorname{arctan}\left(\frac{x}{2}\right)\right) - \frac{3}{2} \operatorname{Ln}\left(1 + \left(\operatorname{arctan}\left(\frac{x}{2}\right)\right)^2\right) + \operatorname{arctan}\left(\operatorname{arctan}\left(\frac{x}{2}\right)\right) + C$$



Método de Integración de irracionales

$$I = \int R(x, x^{\frac{p}{q}}, x^{\frac{r}{s}}, \dots, x^{\frac{u}{v}}) dx \quad I = \int \sqrt{ax^2 + bx + c} dx \quad I = \int \frac{1}{\sqrt{ax^2 + bx + c}} dx$$

Irracionales simples:

$$I = \int \frac{1 + \sqrt{x} + \sqrt[3]{x^2}}{1 + \sqrt[3]{x}} dx \Rightarrow mcm(\text{índices}) = m \Rightarrow x = t^m \quad m = 6$$

$$I = \int \frac{1 + t^3 + t^4}{1 + t^2} (6t^5) dt = 6 \int \frac{t^5 + t^8 + t^9}{1 + t^2} dt = 6 \int t^7 + t^6 - t^5 - t^4 + 2t^3 + t^2 - 2t - 1 + \frac{2t + 1}{1 + t^2} dt$$



Integrales de funciones irracionales

Ejemplo

$$I = \int \frac{2}{[2-x]^2} \sqrt[3]{\frac{2+x}{2-x}} dx \quad t^3 = \frac{2+x}{2-x}$$

$$x = \frac{2t^3 - 2}{1 + t^3} \Rightarrow dx = \frac{6t^2(1 + t^3) - 3t^2(2t^3 - 2)}{[1 + t^3]^2} \Rightarrow dx = \frac{12t^2}{[1 + t^3]^2}$$

$$I = \int \frac{2}{\left[2 - \frac{2t^3 - 2}{1 + t^3}\right]^2} t \frac{12t^2}{[1 + t^3]^2} dt \Rightarrow I = \int 6t^3 dt = \frac{6}{4}t^4 + C$$

$$I = \frac{3}{2} \sqrt[3]{\left[\frac{2+x}{2-x}\right]^4} + C$$



Integrales de funciones irracionales Método del Aleman

$$\int \frac{P(x)}{\sqrt{ax^2 + bx + c}} dx \quad Q(x), \text{ tiene un grado menos que } P(x)$$

$$\frac{P(x)}{\sqrt{ax^2 + bx + c}} = \frac{d}{dx} (Q(x)\sqrt{ax^2 + bx + c}) + \frac{m}{\sqrt{ax^2 + bx + c}}$$

$$\int \frac{P(x)}{\sqrt{ax^2 + bx + c}} dx = (Q(x)\sqrt{ax^2 + bx + c}) + \int \frac{m}{\sqrt{ax^2 + bx + c}} dx$$

Ejemplo

$$I = \int \frac{x^2 - 3x + 7}{\sqrt{2x^2 + 4x + 5}} dx =$$



Integrales de funciones irracionales

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Ejemplo

$$I = \int \frac{x^2 - 3x + 7}{\sqrt{2x^2 + 4x + 5}} dx =$$



Integrales de funciones irracionales

$$I = \int \frac{x^2 - 3x + 7}{\sqrt{2x^2 + 4x + 5}} dx = (Ax + B) \left[\sqrt{2x^2 + 4x + 5} \right] + \int \frac{m}{\sqrt{2x^2 + 4x + 5}}$$

$$\frac{x^2 - 3x + 7}{\sqrt{2x^2 + 4x + 5}} = \frac{d}{dx} (Ax + B) \left[\sqrt{2x^2 + 4x + 5} \right] + \frac{m}{\sqrt{2x^2 + 4x + 5}}$$

$$x^2 - 3x + 7 = A[2x^2 + 4x + 5] + \frac{1}{2}(Ax + B)[4x + 4] + m$$

$$\begin{aligned} A &= \frac{1}{4} \\ B &= \frac{-9}{4} \\ m &= \frac{41}{4} \end{aligned}$$

$$I = \int \frac{x^2 - 3x + 7}{\sqrt{2x^2 + 4x + 5}} dx = \left(\frac{x - 9}{4} \right) \left[\sqrt{2x^2 + 4x + 5} \right] + \frac{41}{4\sqrt{2}} \ln \left[\frac{\sqrt{2}(x + 1) + \sqrt{2x^2 + 4x + 5}}{\sqrt{3}} \right]$$



Integrales de funciones irracionales

Ejemplo

$$I = \int \frac{3x^3 + 2x - 1}{\sqrt{2x^2 + x + 2}} dx = (Ax^2 + Bx + C) \left[\sqrt{2x^2 + x + 2} \right] + \int \frac{m}{\sqrt{2x^2 + x + 2}}$$

$$\frac{3x^3 + 2x - 1}{\sqrt{2x^2 + 4x + 5}} = \frac{d}{dx} (Ax^2 + Bx + C) \left[\sqrt{2x^2 + x + 2} \right] + \frac{m}{\sqrt{2x^2 + x + 2}}$$

$$3x^3 + 2x - 1 = (2Ax + B)[2x^2 + 2x + 2] + \frac{1}{2} (Ax^2 + Bx + C)[4x + 1] + m$$

$$I = \int \frac{x^3 + 2x - 1}{\sqrt{2x^2 + 2x + 2}} dx = \left(\frac{x^2}{2} - \frac{5x}{16} + \frac{15}{64} \right) \left[\sqrt{2x^2 + x + 2} \right] - \frac{63}{128\sqrt{2}} \operatorname{argsenh} \left[\frac{4x + 1}{\sqrt{15}} \right]$$

$$A = \frac{1}{2}$$

$$B = \frac{-5}{16}$$

$$C = \frac{15}{64}$$

$$m = \frac{-63}{128}$$